

Summary of the invariance and procedure of Knockoff

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1 Overview of the Knockoff Filter

The Knockoff filter, originally introduced by Barber and Candès (2015), is a powerful framework for variable selection that provides exact False Discovery Rate (FDR) control in finite samples. Unlike traditional methods that rely on p-values or asymptotic theory, Knockoffs operate by creating counterparts of the original features to serve as a negative control group.

2 The Core Invariance to be guaranteed

The validity of the Knockoff procedure requires specific statistical symmetries throughout the entire pipeline. These symmetries can be categorized into three levels:

2.1 Exchangeability

For a set of covariates $X = (X_1, \dots, X_p)$, we construct knockoff variables $\tilde{X} = (\tilde{X}_1, \dots, \tilde{X}_p)$ fulfilling $\tilde{X} \perp\!\!\!\perp Y|X$. The fundamental requirement is pairwise exchangeability:

$$[X, \tilde{X}] \mid y_{\text{swap}(S)} \stackrel{d}{=} [X, \tilde{X}] \mid y \quad \forall S \subseteq \{1, \dots, p\} \quad (1)$$

where $\text{swap}(S)$ denotes swapping the columns X_j and $\tilde{X}_j$ for all $j \in S$.

2.1.1 Construction of Fixed-X Knockoffs

To ensure the augmented matrix $[X, \tilde{X}]$ is a valid design, its Gram matrix $\mathbf{G}$ must be positive semi-definite. Applying the **Schur Complement** to the block structure yields the critical constraint on the parameter vector s :

$$\mathbf{G} - (\mathbf{G} - \text{diag}(s))\mathbf{G}^{-1}(\mathbf{G} - \text{diag}(s)) \geq 0 \implies \text{diag}(s) \leq 2\mathbf{G} \quad (2)$$

The closed-form solution for the knockoff features is then given by:

$$\tilde{\mathbf{X}} = \mathbf{X}(\mathbf{I} - \mathbf{G}^{-1}\text{diag}(s)) + \mathbf{U}\mathbf{C}, \quad \text{where } \mathbf{C}^\top \mathbf{C} = 2\text{diag}(s) - \text{diag}(s)\mathbf{G}^{-1}\text{diag}(s) \geq 0 \quad (3)$$

Two Strategies for Selecting s

1. **Equi-correlated:** Let $s_j = \min\{1, 2\lambda_{\min}(\mathbf{G})\}$, which is limited by the most correlated pair of variables.

2. **SDP Optimization:** To maximize power, we solve an optimization problem to minimize the pairwise correlation between $\mathbf{X}_j$ and $\tilde{\mathbf{X}}_j$:

$$\min_s \sum_{j=1}^p (1 - s_j) \quad \text{subject to} \quad 0 \leq s_j \leq 1, \quad \text{diag}(s) \leq 2\mathbf{G} \quad (4)$$

By comparison, the **SDP method** is preferred as it avoids the potential problem of the minimum eigenvalue, providing substantially higher power in the presence of complex correlation structures.

2.1.2 Construction of Model-X knockoffs

The Model-X framework extends the knockoff idea to settings where the design matrix X is random and the number of features p may exceed the sample size n .

Covariance Invariance and Gaussian Sufficiency

The construction of Model-X knockoffs relies on the same **Schur Complement** derivation as presented in Section 2.1.1. To ensure a valid joint covariance structure, the parameter vector s must satisfy:

$$\text{diag}(s) \leq 2\Sigma \quad (5)$$

A critical distinction in the Model-X setting is the role of the distribution:

1. **Gaussian Case:** When X follows a multivariate Gaussian distribution, its statistical properties are entirely determined by its mean and covariance. Thus, the second-order constraint above is **sufficient** to guarantee exact distributional exchangeability.
2. **General Case:** For non-Gaussian distributions, this construction ensures *second-order exchangeability*. As emphasized by the author, although higher-order moments might not be perfectly matched, controlling the covariance structure remains highly effective for robust FDR control in practical applications.

ASDP Selection

While the **Equicorrelated** and **SDP** strategies discussed in Section 2.1.1 remain applicable for selecting s in the Model-X framework, they encounter severe computational problem in high-dimensional settings ($p \gg 10^3$).

To address the high complexity of standard SDP, **ASDP (Approximate Semidefinite Programming)** is introduced. ASDP take advantage of the structural properties of Σ by employing sparse approximations. This reduces the computational cost to a near-linear scale, enabling the construction of knockoffs for datasets with hundreds of thousands of variables. Although ASDP provides an approximate solution for s , the resulting loss in statistical power is minimal compared to the gain in computational feasibility, making it the standard choice for large-scale random field models.

2.2 Flip-sign Property

We define a feature importance statistic $W_j = f_j(X, \tilde{X}, Y)$. To ensure the signs of W_j are uninformative for null variables, the function f_j must satisfy:

$$W_j([X, \tilde{X}]_{\text{swap}(j)}, Y) = -W_j([X, \tilde{X}], Y) \quad (6)$$

This ensures that for any j where X_j is a null feature, the distribution of W_j is symmetric around zero, i.e., $W_j \stackrel{d}{=} \epsilon_j |W_j|$, where ϵ_j is an i.i.d. Rademacher random variable.

2.2.1 Construction of Test Statistics: From Fixed-X to Model-X

The knockoff filter performs variable selection by comparing the relative importance of original features and their counterparts. The construction of the test statistic W differs fundamentally between the Fixed-X and Model-X settings.

Statistic under Fixed-X background

In the fixed-X framework, the statistics are typically tied to the Lasso path. Let Z_j and $\tilde{Z}_j$ be the values of the tuning parameter λ at which the j -th feature and its knockoff first enter the Lasso model:

$$Z_j = \sup\{\lambda : \hat{\beta}_j(\lambda) \neq 0\}, \quad \tilde{Z}_j = \sup\{\lambda : \hat{\beta}_{j+p}(\lambda) \neq 0\} \quad (7)$$

The combined statistic is then defined as $W_j = \max(Z_j, \tilde{Z}_j) \cdot \text{sgn}(Z_j - \tilde{Z}_j)$.

Statistic under Model-X background

Model-X knockoffs only lays emphasis on **flip-sign property**. A feature statistic $W = f([X, \tilde{X}], y)$ is valid if and only if swapping X_j with $\tilde{X}_j$ flips the sign of W_j .

- **Feature Importance Measure (Z):** Since the conditional distribution $P(y|X)$ is arbitrary, Z_j can be any valid feature importance measure.
- **Statistic Constructions (W):** There are some examples of statistic construction:
 1. **Difference:** $W_j = Z_j - \tilde{Z}_j$
 2. **Signed Maximum:** $W_j = \max(Z_j, \tilde{Z}_j) \cdot \text{sgn}(Z_j - \tilde{Z}_j)$
 3. **Log-ratio:** $W_j = \log(Z_j/\tilde{Z}_j)$ (if $Z_j, \tilde{Z}_j > 0$)

This flip-sign property ensures that for null features $j \in \mathcal{H}_0$, W_j are i.i.d. symmetric around zero, allowing for exact FDR control via the knockoff threshold.

2.3 Martingale Structure

The final selection is based on a data-dependent threshold T :

$$T = \inf \left\{ t > 0 : \frac{1 + \#\{j : W_j \leq -t\}}{\#\{j : W_j \geq t\} \vee 1} \leq q \right\} \quad (8)$$

This ratio relies on the property that the signs of W_j for null features form a **reverse supermartingale**, allowing for rigorous FDR control even in small samples.